CSHL BIOLOGICAL DATA SCIENCE · NOVEMBER 4-7, 2026 · GALAXY WORKFLOW FOUNDRY

Build better skills. Build better workflows. Keep the receipts.

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A skill is compiled from notes people can read; a workflow is type-checked before any job runs; both print a **receipt you can keep**.

Galaxy PROJECT

Galaxy Workflow Foundry
galaxyproject.github.io/foundry — 69 agent skills compiled from 420 typed notes.

A What it does

The Galaxy Workflow Foundry helps coding agents turn a paper or a Nextflow pipeline into a native, tested Galaxy workflow.

Getting a workflow right is a knowledge problem, not a syntax problem: Galaxy's tools, collections, and idioms change faster than model training data. So the Foundry keeps that knowledge in a wiki people can read and review [3], **compiles it into agent skills**, and lets programs, not the model, decide when a workflow is right. What it can't resolve, it writes down.

Native: Galaxy's own Tool Shed tools, pinned to exact revisions, on Galaxy collections.

Skills are binaries.

KNOWLEDGE BASE
notes, schemas, command manuals, and patterns written for people to read and review

MOLD · ONE ACTION
the exact references one action needs, plus its procedure

↓ *cast: no model, same bytes*

SKILL
frozen, for the agent

PROVENANCE
a hash for every source used

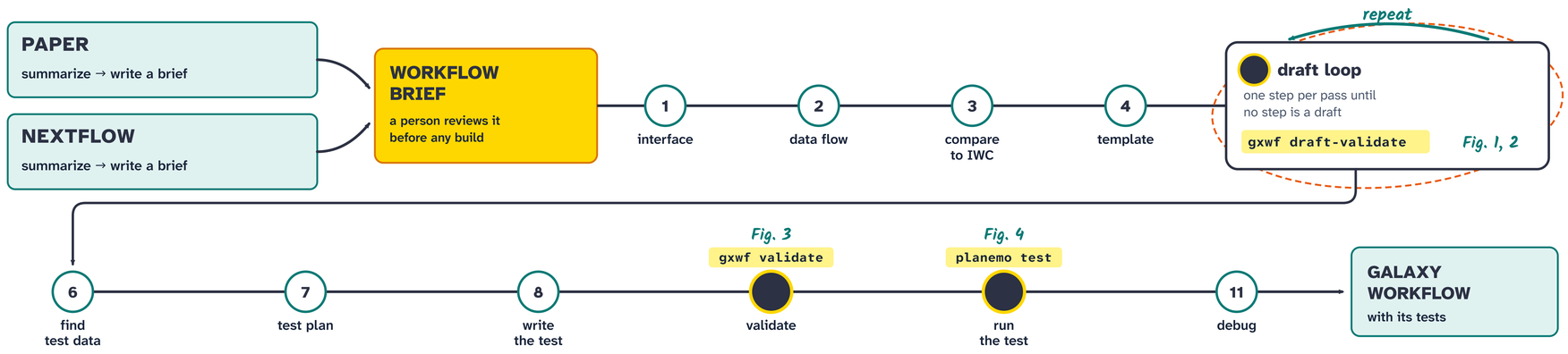
↓ *the agent runs it*

GALAXY WORKFLOW
checked by `gxwf`, run by Planemo

Wrong skill? Fix the note, cast again.

B How it works

A route is a chain of Molds, each cast into a skill. A planning lane turns a paper or a Nextflow pipeline into a **workflow brief**. Once a person has reviewed it, one shared spine of 11 phases builds, checks, and tests the Galaxy workflow. **Filled stations are gates**: a program, not the model, signs each one.



C Inside the loop

Pinned, not named.

Phase 5 builds the workflow one step at a time. Each pass, `gxwf draft-next-step` picks the next unfinished step; the `discover-shed-tool` skill (Fig. 1) searches the Tool Shed and **pins one wrapper to an exact changeset**; the agent fills in the step; `gxwf draft-validate` checks it against that tool's schema. The DengueSeq run took 7 passes.

D Checked, then run

Typed, not guessed.

In phase 9, `gxwf validate` checks every parameter and every connection, collection map-over included, against the tools' real schemas. **No job is queued**, so an agent can repair in a tight loop.

TESTED №4 10-21 00:33
planemo 0.75.47 · its JSON report

planemo test	success
tests	1 · failures 0 · errors 0
Tool Shed repos, at the pins	7
containers	6
wall time	11 min
JOBS RUN	7 ok · 0 failed

SIGNED `planemo test` PASS

Figure 4 · TESTED. Planemo's report of that run. One larger test case (about 178 MB of reads) was not run.

CAST №1 09-13
skill build · foundry @ eb2de3f1

SKILL.md	discover-shed-tool
MOLD rev 5	05c99cba
5 refs	1 upfront · 4 on demand

SIGNED `foundry-build cast`

Figure 1 · CAST. The Tool Shed search skill the DengueSeq run used in its draft loop: Mold revision 5, 5 references, **no model in the build**.

PINNED №2 10-20 22:51–23:39
Tool Shed changesets · phase 5

devteam/bwa_mem	0.7.19+galaxy1 @c598cbb8b019
iuc/samtools_view	1.22+galaxy2 @5cf9fcc93508
iuc/ivar_trim	1.4.4+galaxy1 @6606a8c97889
iuc/ivar_consensus	1.4.4+galaxy0 @1c5448133894
bgruening/tp_sed_tool	9.11+galaxy0 @30b60202d990
iuc/ivar_variants	1.4.4+galaxy0 @5e5dfc9e71e2
iuc/bedtools_genomecoveragebed	2.31.1+galaxy0 @d57471728f7b

SIGNED `gxwf tool-revisions`

Figure 2 · PINNED. The seven tools the run pinned, as `gxwf tool-revisions` reported them. Planemo later installed exactly these.

VALIDATED №3 10-21 00:00
replayed · gxwf 1.15.0 · 2026-10-30

```
$ gxwf validate
  galaxy-workflow.gxwf.yml
  --connections --strict --json
```

Tool state	7 ok · 0 fail · 0 skip
Connections	12 ok · 0 invalid
map-over	all 7 steps over list
wall time	< 1 s

SIGNED `gxwf validate`

Figure 3 · VALIDATED. Phase 9's check, replayed.

E Rails, not reminders

The companion Orbit study ran agents on public Galaxy [1]. Its lesson: **skills are guidance, not enforcement** [1]. What held were the rails. A connection check caught an agent's bad `tximport` wiring before anything ran, and Orbit's evidence gate now refuses by default [1].

The Foundry is built the same way: the model drafts and repairs; `gxwf` and Planemo decide. Its compilation pipeline is described in [1]; here we show its knowledge base, provenance, and checks.

[1] also reports a **60-step workflow** the Foundry compiled from an RNA-seq paper's methods text; comparing with the authors' data found a model choice the methods left out.

What's inside

420 notes · **56** Molds
54 patterns · **69** skills

One note can feed many skills: fix it once, recast, and they all get the fix.

Knowledge-base counts: `galaxyproject/foundry` main, 2026-10-24. Figures 1–4 and the ledger: Foundry run `dengueseq-denv3-galaxy` (2026-10-20), a Galaxy translation of DengueSeq [2]. Times UTC.

[1] Baker D, ..., Nekrutenko A. Verifiable Agentic Science on Public Infrastructure. *bioRxiv* (2026). doi:10.64898/2026.10.26.761864
[2] Vogels CBF, Hill V, et al. DengueSeq. *BMC Genomics* 25:433 (2024).
[3] Karpathy A. LLM wiki. gist.github.com/karpathy (2026).

Gaps are written down.

The run's ledger closed **14 of 19** requirements and left **5 open**, each stated in words for the reviewer: for example, that the source's filtered variant table is not reproduced, a cut the reviewed brief chose.

The last check is a real run.

`gxwf` checks the structure. Then Planemo installs the pinned tools in a real Galaxy and runs the workflow's test. **That run is the validation**: real jobs, real outputs, checked against the expected ones.

F Get started

Claude Code

```
/plugin marketplace add galaxyproject/foundry
/plugin install foundry-skills@galaxy-workflow-foundry
```

Codex

```
codex plugin marketplace add galaxyproject/foundry
codex plugin add foundry-skills@galaxy-workflow-foundry
```

gxwf, the workflow checker

```
npm install -g @galaxy-tool-util/cli
```

From a paper: brief, review it, then build

```
/foundry-skills:pipeline-paper-to-workflow-brief
/foundry-skills:pipeline-workflow-brief-to-galaxy
```

readable wiki ↔ compiled skills **pinned Tool Shed revisions ↔ `gxwf validate`** **real jobs ↔ `planemo test` receipts**

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Foundry docs: galaxyproject.github.io/foundry · Repository: github.com/galaxyproject/foundry · Preprint [1]: doi.org/10.64898/2026.10.26.761864. Related CSHL posters: Galaxy UDTs (van den Beek) · HighGen (Burhans) · disassembler (Seacord).

The Foundry · Preprint [1] · GitHub